

RoundFi Technical Whitepaper

Behavioral Credit Infrastructure for Solana

Version 1.1 — Institutional Revision after PRs 400–405 and Operational Hardening

Executive Summary

RoundFi is a Solana-native behavioral credit protocol. It uses recurring financial participation to create portable on-chain reputation. The protocol starts with savings groups because they create the exact signal DeFi usually lacks: repeated financial discipline under commitment.

The thesis remains simple:

DeFi should not rely only on how much capital a user can lock. It should also evaluate whether the user can behave consistently over time.

RoundFi turns recurring payments into verifiable reputation. That reputation may progressively reduce collateral requirements and eventually become a reusable underwriting signal for DeFi protocols, fintechs, neobanks and credit applications.

The product is not the savings group itself.

The product is the behavioral reputation created by participation.

The current protocol now includes stronger operational maturity, regression protections, canary-stage deployment discipline and bounded economic framing introduced after recent hardening cycles.

1. Core Thesis

RoundFi is based on one core idea:

Recurring financial behavior may become a stronger underwriting signal than static collateral or isolated wallet activity.

Most DeFi lending protocols protect themselves through overcollateralization. This works for market risk, but creates capital inefficiency. Users often need to lock more value than they receive.

RoundFi introduces a different signal: verified recurring behavior.

If a user repeatedly completes financial commitments on-chain, that history may become economically useful over time. RoundFi converts that behavior into portable reputation, then uses that reputation to progressively model lower collateral requirements under bounded conditions.

Importantly, the protocol should not claim that behavioral reputation has already proven capable of

replacing collateral at production scale. That remains an active empirical hypothesis.

2. Problem Statement

DeFi has solved liquidity coordination, automated market making, collateralized lending and settlement infrastructure. It has not fully solved accessible credit at scale.

The current lending model still suffers from three major limitations:

- overcollateralization;
- lack of behavioral underwriting;
- non-portable reputation.

Wallet balances alone do not necessarily represent repayment discipline, long-term coordination quality or recurring reliability.

RoundFi attempts to generate a cleaner behavioral signal directly from recurring participation.

3. What RoundFi Is

RoundFi is a behavioral-credit primitive disguised as a savings protocol.

In Phase 1, users participate in recurring coordination pools.

In Phase 2, successful participation contributes to portable behavioral reputation.

In Phase 3, the reputation layer may evolve into partner-facing infrastructure and APIs.

The savings group is the behavior engine.

The score is the core strategic product.

The oracle/API layer represents the long-term business model.

4. Why Savings Groups

Savings groups naturally generate repeated commitment, peer accountability, recurring participation and scheduled obligations.

RoundFi uses this structure because it produces the exact type of signal credit systems need:

- recurring payments;
- cycle completion;
- late-payment signals;
- default signals;
- participation continuity;
- observable repayment discipline.

The goal is not merely digitizing group savings. The goal is generating reusable underwriting data.

5. Protocol Overview

The protocol combines four major layers:

1. Pool coordination layer.
2. Reputation layer.
3. Yield and reserve layer.
4. Application/indexer layer.

The architecture attempts to transform repeated participation into progressively more useful behavioral coordination signals.

6. Solana-Native Architecture

RoundFi uses Solana because recurring financial participation requires low fees, high throughput, fast settlement and composability.

The architecture currently includes deterministic PDA structures, Anchor programs, SDK infrastructure, wallet flows, Stress Lab simulation, indexer scaffolding and operational hardening systems.

Recent repository evolution also introduced stronger regression philosophy, canary-stage deployment posture and adversarial replay awareness.

7. Account Model

RoundFi relies on deterministic program-derived addresses to maintain auditability and predictable indexing behavior.

Core accounts include protocol configuration, pools, members, reputation profiles, attestations and yield vault state.

Position NFTs remain important because they allow transferable participation positions through the Escape Valve mechanism.

8. User Lifecycle

Users join pools, contribute recurring installments, receive payouts, continue obligations and progressively build behavioral reputation.

The protocol intentionally measures not only whether users receive capital, but whether they continue honoring obligations after receiving liquidity.

9. Reputation Model

RoundFi's reputation model is intentionally behavior-first.

The protocol attempts to evaluate:

- on-time payments;
- completed cycles;
- late payments;
- defaults;
- participation continuity;
- anti-gaming cooldowns;
- optional identity confidence signals.

The protocol avoids presenting the score as a finalized production credit bureau. The current stage should instead be interpreted as a progressively evolving behavioral-signal infrastructure.

10. Dynamic Collateral Reduction

Dynamic collateral reduction remains the protocol's primary economic incentive.

The current modeled ladder remains:

- Level 1 — 50% stake
- Level 2 — 30% stake
- Level 3 — 10% stake

However, these levels should not be interpreted as guaranteed-safe leverage tiers. They represent modeled assumptions under bounded conditions and ongoing behavioral validation.

11. Triple Shield Protection Model

The Triple Shield should now be interpreted as layered mitigation architecture rather than a guarantee system.

The structure combines conservative liquidity behavior, adaptive escrow, member stake exposure, solidarity reserves, deterministic accounting and yield routing.

The updated institutional framing intentionally avoids deterministic “failure-proof” language.

The correct claim is narrower and more rigorous:

RoundFi remained solvent within modeled and tested conditions so far.

12. Yield Waterfall

Yield is not the core thesis of RoundFi.

Behavioral coordination and portable reputation remain the center of the protocol.

Yield exists primarily as an efficiency layer and reserve-support mechanism rather than the foundation of solvency.

The protocol intentionally avoids deterministic profitability or guaranteed APY framing.

13. Guarantee Fund

The Guarantee Fund currently acts as reserve-support infrastructure rather than a universal default guarantee mechanism.

Future catastrophic-loss logic remains roadmap territory and should not be represented as fully shipped functionality.

14. Escape Valve

The Escape Valve allows participants to transfer positions instead of defaulting immediately.

This mechanism attempts to transform some distress scenarios into market-transfer events rather than destructive defaults, preserving coordination continuity and reducing unnecessary reputation damage.

15. Identity Layer

Identity enrichment remains optional and modular.

The protocol is behavior-first rather than identity-first.

Identity may strengthen confidence of positive signals, but participation should not currently depend on mandatory identity verification.

16. Anti-Gaming Design

Any reputation system attracts gaming attempts.

RoundFi attempts to make fake reputation expensive through recurring obligations, cooldowns, economic friction and behavioral continuity requirements.

These mechanisms do not eliminate sybil risk entirely. They attempt to increase the cost of manufacturing high-quality reputation.

17. Stress Lab

The Stress Lab acts as a bounded economic simulator, adversarial replay environment and regression infrastructure.

The simulator models healthy operation, pre-default conditions, post-default scenarios, cascade behavior and veteran-default stress conditions.

Following PRs 400–405, known economic risks increasingly became permanent regression protections rather than remaining theoretical concerns.

The protocol intentionally avoids claiming impossible failure resistance.

18. Implementation Status

The repository currently demonstrates:

- live front-end flows;
- Solana Devnet deployment;
- Stress Lab simulation;
- browser-signed transactions;
- SDK and indexer foundations;
- operational hardening infrastructure;
- open-source repository.

However, Devnet success should not be interpreted as equivalent to production-grade validation.

19. Shipped vs Roadmap

The protocol now maintains stronger separation between shipped functionality, systems under verification and roadmap assumptions.

This distinction is important institutionally because it prevents roadmap claims from being confused with production guarantees.

20. Security Model

Security remains central because the protocol handles financial state and behavioral attestations.

Important areas include PDA validation, authority checks, vault accounting, CPI assumptions, escrow

conditions, NFT transfers and accounting invariants.

External audit remains necessary before meaningful production-scale deployment.

21. Regulatory and Compliance Considerations

RoundFi remains hackathon-stage infrastructure experimentation and should not yet be interpreted as a regulated production credit network.

Future deployment will require jurisdiction-specific legal review, risk controls and compliance evaluation.

22. Key Design Principles

RoundFi continues following several core principles:

- behavior before wealth;
- solvency before growth;
- honest shipped-vs-roadmap separation;
- bounded claims;
- composable infrastructure;
- user benefit before abstraction.

23. Roadmap

The roadmap progresses from prototype validation toward operational hardening, behavioral infrastructure, partner-facing APIs and controlled mainnet deployment.

The current focus increasingly prioritizes empirical behavioral validation rather than theoretical expansion.

24. What RoundFi Is Not

RoundFi is not currently:

- a fully audited production lending protocol;
- a guaranteed-yield system;
- proof that behavioral reputation fully solves credit risk;

- a finalized credit bureau;
- a replacement for legal underwriting.

25. Conclusion

RoundFi attempts to address one of DeFi's hardest problems: reducing reliance on wealth-only underwriting through recurring behavioral coordination data.

The protocol should currently be interpreted as rigorous behavioral-credit infrastructure experimentation under bounded operational and economic assumptions.

26. One-Sentence Summary

RoundFi turns recurring financial behavior into portable on-chain reputation that may progressively reduce collateral friction over time.

27. References

References include repository documentation, architecture specifications, Stress Lab documentation, Devnet deployment records, operational hardening artifacts, SDK infrastructure and economic modeling documentation.